

number of strokes = $0.50 / 0.012 = 42$ (must round up for mark) A1 [3]

3 (a) correct statement, words or symbols B1 [1]

(b)(i) $w = p\Delta V$ C1
 $= 1.03 \times 10^5 \times (2.96 \times 10^{-2} - 1.87 \times 10^{-5})$
 $= (-) 3050 \text{ J}$ A1 [2]

(ii) $q = 4.05 \times 10^4 \text{ J}$ B1 [1]

(iii) $\Delta U = 4.05 \times 10^4 - 3050 = 37500 \text{ J}$...no e.c.f. from (a)..... A1 [1]
 penalise 2 sig.fig. once only

(c) number of molecules = N_A C1
 energy = $37500 / (6.02 \times 10^{23})$
 $= 6.2 \times 10^{-20} \text{ J}$ (accept 1 sig.fig.) A1 [2]